Multi-threaded Web Server Results & Discussion

Made by: Bernardo Reis 126004 | Fernando Santos 124808

In this document, we will present some testing results and our perspective towards our results.

Results of ./full_testing.sh

Our server passed every test (23 in total) that was on the script. The script contained all requested tests in the “Announcement of Project Proposal” PDF file. Here is the output:

```
=== Test: Initialization ===
✓ PASS: Server initializes correctly
=== Test: GET requests for various file type And Content-Type ===
✓ PASS: GET and Content-Type works for PNG
✓ PASS: GET and Content-Type works for PDF
✓ PASS: GET and Content-Type works for HTML
✓ PASS: GET and Content-Type works for JS
✓ PASS: GET and Content-Type works for CSS
=== Test: Verify correct HTTP status codes ===
✓ PASS: Status 200
✓ PASS: Status 404
✓ PASS: Status 401
✓ PASS: Status 403
✓ PASS: Status 500
✓ PASS: Status 503
=== Test: Directory index serving ===
✓ PASS: Directory listing
=== Test: Concurrent requests ===
Testing concurrent requests! (This may take a while...)
Completed 1000 requests
Completed 2000 requests
Completed 3000 requests
Completed 4000 requests
Completed 5000 requests
Completed 6000 requests
Completed 7000 requests
Completed 8000 requests
Completed 9000 requests
Completed 10000 requests
Finished 10000 requests
✓ PASS: No Concurrent Requests Dropped
✓ PASS: Statistics accuracy under concurrent load
=== Test: Multiple parallel curl clients ===
✓ PASS: Parallel clients (20/20 succeeded)
=== Test: Log file integrity (no interleaved entries) ===
✓ PASS: Log entries are not interleaved (50/50 complete)
=== Test: Cache consistency across threads ===
✓ PASS: Cache consistent across threads (30 requests, 1 unique content)
=== Test: Helgrind ===
✓ PASS: Helgrind
=== Test: Valgrind ===
✓ PASS: Valgrind
=== Test: Graceful Shutdown ===
✓ PASS: Graceful shutdown
=== Test: Zombie Processes ===
✓ PASS: No Zombie processes (0 found)
=== Test: Continuous Load (5+ minutes) ===
⚠ WARNING: This test will run for 5 minutes with continuous load.
Do you want to continue? (y/N): y
Starting continuous load test at sex 12 dez 2025 21:34:21 WET
Running Apache Bench for 5 minutes with 50 concurrent connections...
Continuous load test completed at sex 12 dez 2025 21:39:23 WET
Actual duration: 300 seconds
Total requests: 340000
Failed requests: 0
✓ PASS: Continuous load (340000 requests, 0 failed, 300s)
========================================
Results: 23 passed, 0 failed
========================================
```

Apache Bench Results

With ab –n 10 https://localhost:8080/:

Bernardo Reis | 126004
Fernando Santos | 124808

```
Server Software:        ConcurrentHTTP/1.0
Server Hostname:        localhost
Server Port:            8080
SSL/TLS Protocol:       TLSv1.3,TLS_AES_256_GCM_SHA384,4096,256
Server Temp Key:        X25519 253 bits
TLS Server Name:        localhost

Document Path:          /
Document Length:        42551 bytes

Concurrency Level:      1
Time taken for tests:   0.511 seconds
Complete requests:      100
Failed requests:        0
Total transferred:      4266500 bytes
HTML transferred:       4255100 bytes
Requests per second:    195.56 [#/sec] (mean)
Time per request:       5.113 [ms] (mean)
Time per request:       5.113 [ms] (mean, across all concurrent requests)
Transfer rate:          8148.14 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        3    5   1.4      4      15
Processing:     0    0   0.1      0       1
Waiting:        0    0   0.1      0       1
Total:          4    5   1.5      5      15

Percentage of the requests served within a certain time (ms)
  50%      5
  66%      5
  75%      5
  80%      5
  90%      6
  95%      8
  98%     10
  99%     15
 100%     15 (longest request)
```

With ab –n 10000 -k https://localhost:8080/:

Bernardo Reis | 126004
Fernando Santos | 124808

```
Server Software:        ConcurrentHTTP/1.0
Server Hostname:        localhost
Server Port:            8080
SSL/TLS Protocol:       TLSv1.3,TLS_AES_256_GCM_SHA384,4096,256
Server Temp Key:        X25519 253 bits
TLS Server Name:        localhost

Document Path:          /
Document Length:        42551 bytes

Concurrency Level:      1
Time taken for tests:   48.509 seconds
Complete requests:      10000
Failed requests:        0
Keep-Alive requests:    0
Total transferred:      426650000 bytes
HTML transferred:       425510000 bytes
Requests per second:    206.15 [#/sec] (mean)
Time per request:       4.851 [ms] (mean)
Time per request:       4.851 [ms] (mean, across all concurrent requests)
Transfer rate:          8589.10 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        3    5   0.8      4      13
Processing:     0    0   0.1      0       1
Waiting:        0    0   0.1      0       1
Total:          3    5   0.8      5      14
WARNING: The median and mean for the initial connection time are not within a normal deviation
        These results are probably not that reliable.

Percentage of the requests served within a certain time (ms)
  50%      5
  66%      5
  75%      5
  80%      5
  90%      6
  95%      6
  98%      7
  99%      8
 100%     14 (longest request)
```

With ab –n 1000 –c 10 https://localhost:8080/:

Bernardo Reis | 126004
Fernando Santos | 124808

```
Server Software:        ConcurrentHTTP/1.0
Server Hostname:        localhost
Server Port:            8080
SSL/TLS Protocol:       TLSv1.3,TLS_AES_256_GCM_SHA384,4096,256
Server Temp Key:        X25519 253 bits
TLS Server Name:        localhost

Document Path:          /
Document Length:        42551 bytes

Concurrency Level:      10
Time taken for tests:   0.915 seconds
Complete requests:      1000
Failed requests:        0
Total transferred:      42665000 bytes
HTML transferred:       42551000 bytes
Requests per second:    1092.35 [#/sec] (mean)
Time per request:       9.155 [ms] (mean)
Time per request:       0.915 [ms] (mean, across all concurrent requests)
Transfer rate:          45512.83 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        4    8   2.9      7      18
Processing:     0    1   1.1      1       7
Waiting:        0    0   0.4      0       5
Total:          4    9   3.3      8      22

Percentage of the requests served within a certain time (ms)
  50%      8
  66%      9
  75%     10
  80%     11
  90%     14
  95%     16
  98%     18
  99%     19
 100%     22 (longest request)
```

With ab -n 10000 -c 200 https://localhost:8080/:

Bernardo Reis | 126004
Fernando Santos | 124808

```
Server Software:        ConcurrentHTTP/1.0
Server Hostname:        localhost
Server Port:            8080
SSL/TLS Protocol:       TLSv1.3,TLS_AES_256_GCM_SHA384,4096,256
Server Temp Key:        X25519 253 bits
TLS Server Name:        localhost

Document Path:          /
Document Length:        42551 bytes

Concurrency Level:      200
Time taken for tests:   7.449 seconds
Complete requests:      10000
Failed requests:        0
Total transferred:      426650000 bytes
HTML transferred:       425510000 bytes
Requests per second:    1342.49 [#/sec] (mean)
Time per request:       148.977 [ms] (mean)
Time per request:       0.745 [ms] (mean, across all concurrent requests)
Transfer rate:          55935.00 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        5  108  61.0     93     327
Processing:     0   39  30.0     29     148
Waiting:        0    4   4.3      3      33
Total:         31  147  51.5    135     344

Percentage of the requests served within a certain time (ms)
  50%    135
  66%    149
  75%    163
  80%    175
  90%    216
  95%    264
  98%    306
  99%    320
 100%    344 (longest request)
```

With the extreme scenario ab -n 20000 -c 500 –k https://localhost:8080/:

Bernardo Reis | 126004
Fernando Santos | 124808

```
Server Software:        ConcurrentHTTP/1.0
Server Hostname:        localhost
Server Port:            8080
SSL/TLS Protocol:       TLSv1.3,TLS_AES_256_GCM_SHA384,4096,256
Server Temp Key:        X25519 253 bits
TLS Server Name:        localhost

Document Path:          /
Document Length:        42551 bytes

Concurrency Level:      500
Time taken for tests:   12.676 seconds
Complete requests:      20000
Failed requests:        0
Keep-Alive requests:    0
Total transferred:      853300000 bytes
HTML transferred:       851020000 bytes
Requests per second:    1577.73 [#/sec] (mean)
Time per request:       316.911 [ms] (mean)
Time per request:       0.634 [ms] (mean, across all concurrent requests)
Transfer rate:          65736.25 [Kbytes/sec] received

Connection Times (ms)
              min  mean[+/-sd] median   max
Connect:        5  187  53.2    178     414
Processing:    23  127  22.2    121     198
Waiting:        0   11   5.6     10      39
Total:        136  315  59.1    307     589

Percentage of the requests served within a certain time (ms)
  50%    307
  66%    327
  75%    339
  80%    348
  90%    397
  95%    441
  98%    473
  99%    496
 100%    589 (longest request)
```

Discussion

We are proud of our server performance, and its ability to handle a large number of requests, with HTTPS in a short time. Our biggest difficulty was the handling of keep-alive requests, since we were not distinguishing keep-alive from non-keep-alive connections. However, after some changes, this problem was finally solved. This breakthrough was a major step towards our results.

Bernardo Reis | 126004
Fernando Santos | 124808